

Article

Metal Acetate-Enhanced Microwave Pyrolysis of Waste Textiles for Efficient Syngas Production

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Abstract: The production of waste textiles has increased rapidly in the past two decades along with the rapid development of the economy, the majority of which has been either landfilled or incinerated, resulting in energy loss and environmental pollution. Microwave pyrolysis, which can transform heterogeneous and complex waste feedstocks into value-added products, is considered one of the most competitive technologies for processing waste textiles. However, achieving selective product formation during the microwave pyrolysis of waste textiles remains a significant challenge. Herein, sodium acetate, potassium acetate, and nickel acetate were introduced into waste textiles through an impregnation method as raw materials to improve the pyrolysis efficiency. The optimized process parameters indicated that nickel acetate had the most favorable promotional effect of the three acetates. Notably, the waste textiles containing 1.0% Ni exhibited the highest gas production rate, with the hydrogen-containing combustible gas reaching 81.1% and 61.0%, respectively. Using X-ray diffraction (XRD), transmission electron microscopy (TEM), Fourier transform infrared spectroscopy (FTIR), and Raman spectroscopy to characterize the waste textiles before and after pyrolysis, it was found that nickel acetate was converted into metallic nickel (Ni⁰) during microwave pyrolysis. This active site significantly enhanced the pyrolysis process, and as the gas yield increased, the disorder of the resulting pyrolytic carbon also rose. The proposed Ni⁰-enhanced microwave pyrolysis mediated by nickel acetate offers a novel method for the efficient disposal and simultaneous resource recovery of waste textiles.

Keywords: waste textiles; metal acetate; microwave pyrolysis; zero-valent nickel; syngas



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1. Introduction

The rapid industrialization in recent years has resulted in a significant increase in the generation of solid waste [1–3]. Waste textiles are a common form of organic solid waste, with more than 92 million tons generated globally each year [4]. Currently, only about 15% of waste textiles are recycled, while most of them are dumped in landfills or incinerated [5]. Landfills are now rarely used due to the high land resource occupation and the high risk of secondary pollution [6]. Incineration is highly effective in reducing the solid waste volume, but its energy recovery remains limited, and resource utilization is relatively inefficient. Moreover, the incineration process easily produces highly toxic, carcinogenic substances such as dioxins, leading to secondary environmental pollution and posing a new challenge in the context of carbon neutrality [7,8]. Therefore, it is crucial to develop green and sustainable technologies for the resource recovery of waste textiles.

Pyrolysis, a thermal decomposition process in which organic matter is decomposed into pyrolysis gas, oil, and carbon by high-temperature heat treatment in the absence/lack of oxygen, is widely regarded by researchers as an effective method for treating organic solid waste due to its environmental benefits and potential for efficient resource utilization [9–11]. However, traditional pyrolysis driven by electric heating faces several limitations, including slow heating rates, prolonged processing times, and low yields of pyrolysis gas. Among the various branches of pyrolysis, microwave pyrolysis initiated by electromagnetic waves has emerged as a leading alternative, offering significant improvements over conventional methods. Microwave pyrolysis is characterized by its rapid and uniform heating, high energy efficiency, and ease of operational control [12]. These advantages have made it a focal point of increasing interest in waste treatment technologies. Unlike conventional pyrolysis, which relies on external heat sources and can lead to uneven temperature distribution, microwave pyrolysis directly heats the material at the molecular level through electromagnetic waves, resulting in more efficient and controlled thermal decomposition [13–15]. Parvez et al. studied and compared the conventional and microwave-assisted biomass pyrolysis processes at the laboratory scale and found that microwave pyrolysis has a higher gas yield than that of conventional pyrolysis, and the energy and exergy rates of the pyrolysis gas in the microwave process is higher than that in the conventional process [16]. Kim et al. conducted a study on the sludge treatment of sludge and found that microwave pyrolysis is a very efficient process [17]. They used both conventional and microwave heating for hydrogen production from sludge, and found that microwave heating was most efficient at 800 °C. The energy consumption of the microwave during the process of rising from an ambient to the optimal temperature and maintaining it was lower than that of the conventional method, due to the better wave-absorbing properties of the biocarbon produced by the pyrolysis and the shorter heating time.

Research on obtaining valuable substances from solid wastes through pyrolysis technology is in full swing. However, most of the work focuses on increasing the oil yield from pyrolysis, while the complexity and poor stability of pyrolysis oil and its narrow downstream application range limit the promotion of pyrolysis technology [18,19]. On the contrary, the pyrolysis technology route, which focuses on the production of hydrogen-containing combustible gas (H_2 , CO, CH_4 , C_2H_4 , and C_2H_6), has received extensive attention from researchers due to the great potential of hydrogen application in the future energy system, as well as the important use of CO, CH_4 , C_2H_4 , and C_2H_6 in the field of chemistry and chemical industry, which demonstrates excellent practicability and economic performance [20,21]. In the pyrolysis process, gas-phase small molecules are mainly produced by the fracture of C-C and C-H in large molecules, and it has been shown that as many gas-phase small molecules as possible can be produced by increasing the pyrolysis temperature or prolonging the reaction time [22,23]. However, increasing the temperature or extending the duration means more energy input to the system, with poor economics. During the pyrolysis reaction, the addition of a catalyst (such as Fe) can reduce the activation energy of the reaction, which enables the controlled regulation of the product distribution under relatively mild conditions in order to obtain as many gas-phase products as possible [24–26]. In the field of biomass pyrolysis, a great deal of work has been carried out by researchers regarding the effect of intrinsic or exogenously introduced metal salts on the distribution of pyrolysis products. Patwardhan et al. found that alkali metals in the biomass could promote the secondary pyrolysis reaction of the products and increase the proportion of gas yields [27]. Shang et al. found that metal salt could reduce the pyrolysis liquid phase product yield and increase the gas-phase product yield [28]. For waste textiles, Yousef et al. found that metal ions in denim had an effect on the distribution of the pyrolysis products [29]. It could be seen that the introduction of metal salts into the pyrolysis system positively contributed to the regulation of the pyrolysis products, which could effectively improve the yield of the gas-phase products.

Herein, we innovatively employ the impregnation method to load sodium acetate, potassium acetate, and nickel acetate into waste textiles as raw materials for in situ catalytic

pyrolysis in a microwave pyrolysis reactor. The influence of different metal salt species on the pyrolysis behavior of the textiles is investigated, identifying the metal salts that yield the highest amount of gas and hydrogen-rich combustible gases. Further optimization of the selected metal salt content enhances the performance, leading to the establishment of optimal conditions for the catalytic pyrolysis of waste textiles. A comprehensive analysis of the textiles before and after pyrolysis is conducted using thermogravimetric analysis (TG), X-ray diffraction (XRD), Fourier transform infrared spectroscopy (FTIR), and Raman spectroscopy. Integrating these characterization techniques with pyrolysis evaluation data allows for the deduction of the active center responsible for metal salt catalysis. Additionally, the performance data of conventional pyrolysis are compared with those of microwave pyrolysis for treating waste textiles, highlighting the advantages of the microwave approach. This study presents a novel strategy for enhancing the efficiency of waste textile treatment through targeted catalytic pyrolysis.

2. Experiment

2.1. Materials

Waste textiles (Figure S1) were sourced from Guangdong Province, China. The waste textiles were cut into small squares (4 mm × 4 mm), and the pieces were dried in an oven at 110 °C for 12 h before the tests. The element analysis was conducted in a Vario MACRO cube Elemental analyzer (Germany) and the results of element components and element percentage of waste textile are displayed in Table S1. $\text{CH}_3\text{COONa} \cdot 3\text{H}_2\text{O}$ (99.5%), CH_3COOK (99.0%), and $\text{Ni}(\text{CH}_3\text{COO})_2 \cdot 4\text{H}_2\text{O}$ (99.0%) were analytical-grade, purchased from Shanghai Aladdin Biochemical Technology Co., Ltd. (Shanghai, China). The metal-containing waste textile was prepared via an incipient wetness impregnation method [30,31]. An appropriate amount of metal acetate was dissolved in 6.50 mL deionized water, then 2.50 g waste textile was added into the obtained solution. After impregnation, the mixture was dried at 110 °C for 16 h. The waste textiles impregnated with CH_3COONa and CH_3COOK were prepared, and the contents of sodium and potassium were 2.6 wt.%, which were labeled as 2.6Na and 2.6K, respectively. The metal content of the sample was tested by inductively coupled plasma-optical emission spectrometry (ICP-OES, Agilent ICP730, Agilent, California, USA). Meanwhile, a series of $\text{Ni}(\text{CH}_3\text{COO})_2$ -containing waste textiles with nickel contents of 0.7, 1.0, 2.6, 3.5, and 6.1 wt.% were obtained, denoted as 0.7Ni, 1.0Ni, 2.6Ni, 3.5Ni, and 6.1Ni, respectively.

2.2. Experimental Apparatuses and Pyrolysis Performance Evaluation

The pyrolysis tests of obtained materials were performed in a microwave pyrolysis furnace (CY-PY1100C-S, Changyi, Changsha, China) with a microwave frequency of 2450 ± 50 MHz and microwave power of 1.4 kW, which provides a controlled atmosphere and high-temperature environment. The experimental device for microwave pyrolysis is presented in Figure 1. To guarantee an inert reaction environment, nitrogen gas (N_2 , 99.9%) was continuously introduced into the quartz reactor for 15 min at a flow rate of 100 mL/min ahead the pyrolysis. In a typical run, 1.0 g of the as-prepared material was loaded into a SiC boat with significant microwave absorbing ability. The sample was heated to 500 °C in the microwave pyrolysis furnace, and maintained for 5 min at a N_2 flow rate of 40 mL/min. The microwave power was adjusted automatically according to the displayed temperature to maintain the final temperature of the pyrolysis at a constant. For comparison, conventional pyrolysis was performed at 500 °C for 5 min. The gaseous product was cooled by the condenser and collected in a gas bag made of PTFE for further analysis. The pyrolysis residues were cooled to room temperature and collected from the SiC boat, and then weighed. This study focused on gaseous products which were analyzed using a gas chromatograph (GC9790 Plus, FULI Instruments, Wenling, China) equipped with a TCD detector. The operation conditions were set as follows: carrier gas was Argon (Ar) and the flow rate was 20.0 mL/min; injection volume was 1 mL; detector temperature operated at 120 °C, the temperature of thermal conductivity cell was set at 120 °C and the

column oven temperature was 70 °C. Each experiment was repeated three times, and the average value was taken to ensure the reliability of the data.

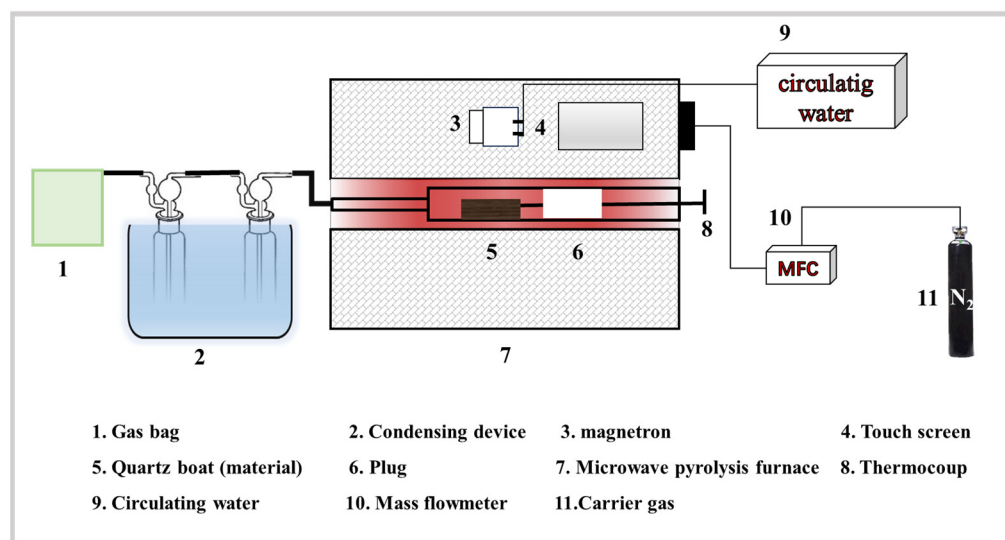


Figure 1. Schematic of microwave pyrolysis system.

To evaluate the catalytic performance, some parameters are defined. The solid, liquid, and gas yield are calculated as follows:

$$\text{Solid yield (\%)} = \frac{m_{\text{solid}}}{m_{\text{obtained materials}}} \times 100\% \quad (1)$$

$$\text{Liquid yield (\%)} = \frac{m_{\text{liquid}}}{m_{\text{obtained materials}}} \times 100\% \quad (2)$$

$$\text{Gas yield (\%)} = \frac{m_{\text{obtained materials}} - m_{\text{solid}} - m_{\text{liquid}}}{m_{\text{obtained materials}}} \times 100\% \quad (3)$$

where $m_{\text{obtained materials}}$ (g) is the mass of raw or metal-containing waste textiles, m_{solid} (g) is the mass of residue solid after the microwave pyrolysis test, and m_{liquid} (g) is the mass of liquid collected during the microwave pyrolysis test.

Pyrolysis performance is characterized by the gas concentration and yield. The gas concentration (C_i , vol.%) is the volume percentage of the gas composition in the syngas, and in this study, the hydrogen concentration is mainly used. The gas concentration is calculated as follows:

$$C_i = \frac{x_i}{x_{H_2} + x_{CO} + x_{CO_2} + x_{CH_4} + x_{C_2H_m}} \quad (4)$$

where x_i (vol.%) is the percentage of gas volume detected by GC.

2.3. Characterization

The crystal structure of the catalyst was analyzed by an X-ray diffraction analyzer (XRD, BRUKER, D8 ADVANCE, Karlsruhe, Germany). The microstructure and element mapping of catalysts were characterized by scanning electron microscopy (SEM, Quanta650, Hitachi, Tokyo, Japan). A HORIBA LabRAM HR Evolution Laser Raman spectrometer (HORIBA JY, Paris, France) was used to test the prepared samples. According to the experimental requirements, a UV 325 nm light source, grating with 2400 grooves mm^{-1} , and CCD camera were selected for the detector. Raman spectra of 100–800 cm^{-1} were recorded at room temperature. Fourier transform infrared spectroscopy (FT-IR) was conducted to analyze the chemical bonds and functional groups of the samples by using the SHIMADZU IRTracer-100 (Shimadzu, Kyoto City, Japan). A powered sample (ca. 1 mg but precisely

weighed) and KBr (ca. 200 mg but precisely weighed) was mixed and ground uniformly. The mixture was pretreated at 60 °C in a vacuum oven for 1 h to remove physical water from the catalyst. The dried sample was pressed into a tablet and measured by FT-IR. A series of TG (NETZSCH STA 449 F3, Netzsch, Selb, Bavaria, Germany) tests were performed to understand the thermolytic characteristics of waste textiles under N₂ atmospheric conditions. TG experiments of waste textiles (10.00 ± 0.05 mg) were conducted from 40 to 900 °C with a constant heating rate (10 °C/min).

3. Results and Discussion

3.1. Optimization of Metal Acetate Types and Contents for Microwave Pyrolysis of Waste Textiles

To investigate the performance of the microwave pyrolysis of waste textiles to produce hydrogen-containing gas, microwave pyrolysis tests were carried out on metal-containing waste textiles impregnated with different types and contents. Figure 2a represents the three-phase product distribution obtained during the microwave pyrolysis of the waste textiles, Na-containing waste textiles, K-containing waste textiles, and Ni-containing waste textiles. The results showed that the highest gas yield (78.6%) was obtained from the microwave pyrolysis experiments using the Ni-containing waste textile compared to the original waste textile, the K-containing waste textile had no effect on the distribution of the three-phase products during the microwave pyrolysis, but the Na-containing waste textile was not favorable for microwave pyrolysis to prepare the gas-phase products, and its gas content was reduced (8.5%). The results of further analysis of the composition of the gas products obtained during microwave pyrolysis are shown in Figure 2b. The presence of metal salts enhanced the content of combustible gas and H₂, and, compared with the K-containing and Na-containing waste textiles, the Ni-containing waste textiles exhibited more significant catalytic promotion of H₂ production, which could substantially increase the hydrogen content of the gas products. The H₂ content increased from 0.5% of the waste textiles to 16.1%, which was a 32.2-fold enhancement.

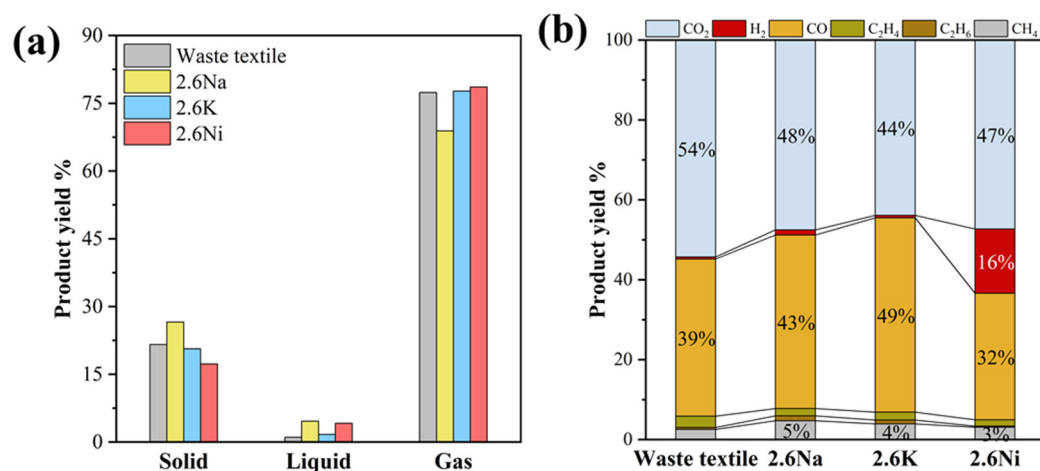


Figure 2. (a) Effect of metal salt species on the catalysis yield of three-phase products of textiles and (b) effect of metal salt species on the catalysis yield of gas-phase products of textiles.

Different contents of Ni salt were impregnated onto the waste textiles to investigate the constitutive relationship between Ni and the waste textiles during pyrolysis as well as the optimal Ni loading. As shown in Figure 3a, the microwave pyrolysis of the 1.0 wt.% Ni-impregnated waste textile showed the highest gas yield (81.1%), which was 4.8% higher than that of the original waste textile. However, with the further increase in the Ni content, the gas yield decreased and the liquid yield increased, which was presumed to be due to the aggregation of some of the Ni caused by the increase in the Ni content, resulting in a decrease in active sites. As shown in Figure 3b, the catalytic pyrolysis of hydrogen-containing combustible gas (H₂, CO, CH₄, C₂H₄, and C₂H₆) was the best when impregnated with 1.0% Ni, and the content of hydrogen-containing combustible gas produced by the

microwave pyrolysis of the waste textiles gradually decreased after impregnating with more than 1.0% Ni. The 1.0% Ni waste textile corresponded to a combustible gas yield of up to 61.0%, and the solid and liquid products of the microwave pyrolysis of the 1.0% Ni-impregnated feedstock to make gas were less, at only 15.7% and 3.2%. The pyrolysis reaction was more adequate, which effectively promoted the pyrolysis of the waste textiles to prepare combustible gas and reduced the yield of biochar and bio-oil.

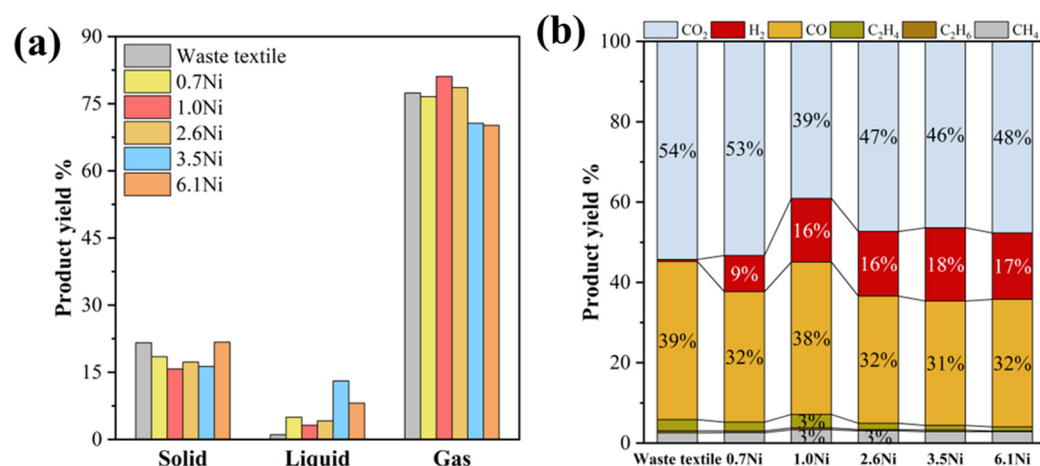


Figure 3. (a) Effect of Ni metal salt addition amount on the catalysis yield of three-phase products; (b) effect of Ni metal salt addition amount on the catalysis yield of gas-phase products.

We further investigated the microwave pyrolysis results with FTIR and Raman tests. Figure 4 showed the FTIR results of the obtained materials before and after the pyrolysis tests. As shown in Figure 4a, an absorption peak with a higher intensity at 3426 cm^{-1} was attributed to the O-H stretching vibration of the metal salt-containing waste textile [32]. The characteristic peak at 2893 cm^{-1} was attributed to the C-H bending vibration of the methyl group in the waste textile, and the characteristic peaks at $1380\sim 1600\text{ cm}^{-1}$ were attributed to the C-O stretching vibration of acetate ions [33]. The characteristic peak at 1045 cm^{-1} was attributed to the bending vibration of C-O in the acetate ion or waste textile [32,33]. The peak at 660 cm^{-1} corresponded to the stretching vibration of M-O, which further confirmed that the structure of acetate did not change during the process of acetate impregnation in the shredded textile preparation [34]. Figure 4b showed the FTIR results of the obtained materials after the pyrolysis tests, and it was clearly observed that the characteristic peaks of C-H and C-O markedly reduced after the microwave reaction, which indicated that the waste textiles had been pyrolyzed. Ni-containing waste textile has a better fabric cracking ability because it has the lowest intensity of the bending vibration of C-H and C-O in the waste textile compared to the other metal-containing waste textiles after microwave pyrolysis.

The Raman spectra of the obtained materials after the pyrolysis test were obtained out and the results are shown in Figure 5. Typical D and G peaks of carbon materials can be observed around 1355 and 1585 cm^{-1} . The D peak was caused by defects in the graphite layer, which was present in the graphite-like and other forms of carbonaceous materials, and was attributed to the boundary vibrational mode of the disorder-induced hexagonal Brillouin zone around 1350 cm^{-1} , which was commonly used for defect characterization. The G peak was caused by the sp^2 carbon atoms between the graphite layers around 1580 cm^{-1} , which was attributed to the stretching vibrational mode of the in-plane bonding of carbon atoms, and was related to the degree of graphitization [35,36]. Its peak width and peak intensity were also related to the defects. The peak position is also related to the morphology of the carbon material. The intensity ratio of the D and G bands (ID/IG) of the carbon material is usually related to the degree of structural disorder and the number of defects. The higher the ratio, the more bonding disorder, vacancies, and defects. Figure 5a shows the Raman spectra of the waste textiles containing different kinds of metal salts

after the microwave pyrolysis. It can be seen that the ID/IG (Table 1) of the solid after the microwave pyrolysis of the Na-containing waste textile is lower, even lower than that of the original waste textile, which indicated that the Na metal was unfavorable to the pyrolysis of the waste textile. This was in agreement with the results of the performance evaluation (Figure 2). The K-containing waste textile after microwave pyrolysis had the highest ID/IG, and its microwave pyrolysis gas-phase product contained a high content of combustible gas, but it was not the optimal metal salt because of its very low hydrogen content (1.0%). The ID/IG of the solid-phase product after the microwave pyrolysis of the Ni-containing waste textile was 0.7862, and its hydrogen content in the gas reached 16.0%, which was much better than that for the other metal salts. The ID/IG ratio gradually decreased with the increase in the impregnated Ni content (Figure 5b), indicating that too high a Ni content was not favorable to the pyrolysis of the waste textiles, which was consistent with the experimental results (Figure 3). According to the above discussion, Ni has an excellent syngas yield compared to that of K and Na. Alkali metals (Na and K) tend to favor the production of char, which results in a relatively high amount of solid-phase product [37]. As shown in Figure 2a, it could be inferred that the waste textile impregnated with Na would produce more solid-phase products than the pristine waste textile. However, the solid-phase product of the waste textile impregnated with K changed slightly, which corresponded to the reduction in the thermal decomposition temperature of biomass by adding K and the prolongation of the duration of the pyrolysis process [38]. Ni species have been widely adopted as active metals for the steam reforming of oxygenates due to their high activity of C-C/C-H bond cleavage and H₂ formation from H atoms [39,40], which account for the superior syngas production.

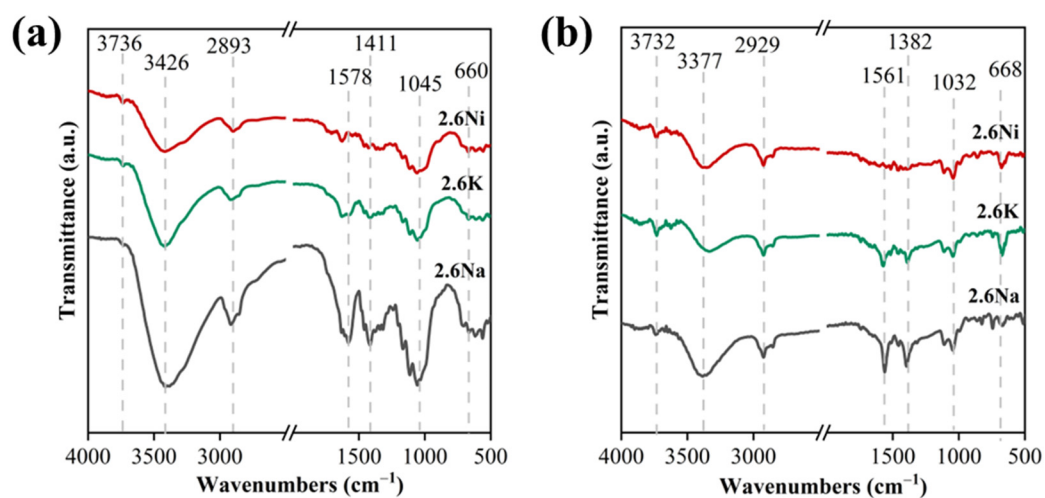


Figure 4. (a) FTIR of original waste textiles and waste textiles impregnated with different metal salts before microwave pyrolysis. (b) FTIR of original waste textiles and waste textiles impregnated with different metal salts after microwave pyrolysis.

Table 1. The intensity ratio of the D and G bands (ID/IG) of the carbon material after microwave pyrolysis calculated from the Raman results.

Materials	Waste Textile	2.6Na	2.6K	1.0Ni	2.6Ni	3.5Ni	6.1Ni
ID/IG	0.7198	0.6498	0.8226	0.7862	0.7296	0.6970	0.6293

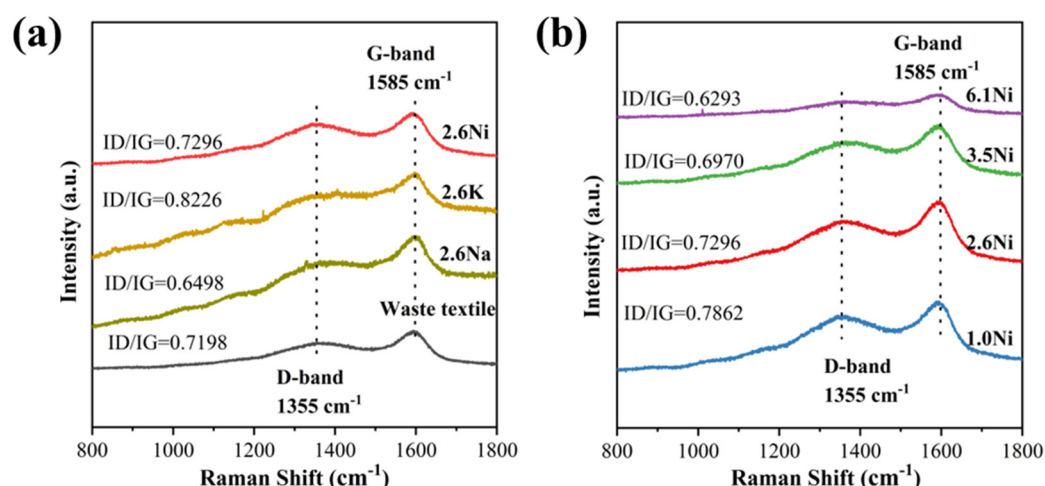


Figure 5. (a) Raman spectrum of original waste textiles and waste textiles impregnated with different metal salts after microwave pyrolysis. (b) Raman spectrum of waste textiles impregnated with different Ni concentrations after microwave pyrolysis.

3.2. Zero-Valent Nickel-Enhanced Microwave Pyrolysis of Waste Textiles to Produce Syngas

In order to investigate the role of Ni in the preparation of hydrogen-containing combustible gas by the microwave pyrolysis of waste textiles, the physical and chemical properties of the samples before and after the microwave pyrolysis of the original waste textile and Ni-containing waste textile were characterized and analyzed. The XRD curves (Figure 6a) showed that when 1.0% Ni was loaded onto the waste textile, the Ni salt did not affect the structure of the waste textile, and there were no characteristic peaks clearly attributed to the Ni species, indicating that Ni was uniformly dispersed on the surface of the waste textile and did not agglomerate. Further analysis of the reacted waste textile and Ni-containing waste textile showed that the reacted waste textile had one obvious carbon peak at 23°, and the peaks at 45° and 50° assigned to Ni⁰ were observed in the reacted Ni-containing waste textile [41,42]. It could be concluded that Ni species have the best C-C and C-H bond-breaking ability in their metal form during the microwave pyrolysis tests. Smooth structures were observed by SEM of the waste textile and 1.0Ni waste textile (Figure 6b,c), indicating that the load of Ni does not affect the surface structure of the textile. No particle species were observed, which further indicates that the introduced Ni is evenly distributed on the textile. After microwave pyrolysis, a rough surface and small components were observed on the 1.0Ni-AF samples (Figure 6d), suggesting that the textile structure was destroyed. The results of the SEM mapping analysis (Figure S2) suggested that Ni was evenly distributed on the textile before and after the reaction.

We further investigated the state of nickel acetate before and after microwave pyrolysis at 500 °C by FTIR. Figure S3 shows the IR spectra of the dried nickel acetate, in which the characteristic peaks in the band at 1416–1594 cm⁻¹ were attributed to the symmetric stretching vibration of the C-O double bond of the acetate ion. The characteristic peaks at the band at 3416 cm⁻¹ were attributed to the stretching vibration of O-H in the acetate ion, and the characteristic peak at 1025 cm⁻¹ was attributed to the bending vibration of C-O in the acetate ion. The peak at 685 cm⁻¹ corresponded to the metal–oxygen bonding of Ni-O. After microwave pyrolysis at 500 °C, there were no obvious characteristic peaks of nickel acetate. Combined with the analysis of the XRD spectra, it can be seen that the acetate ions in nickel acetate were decomposed completely during microwave pyrolysis, and the nickel ions were converted into Ni⁰, which then participated in the pyrolysis of the waste material as the active sites.

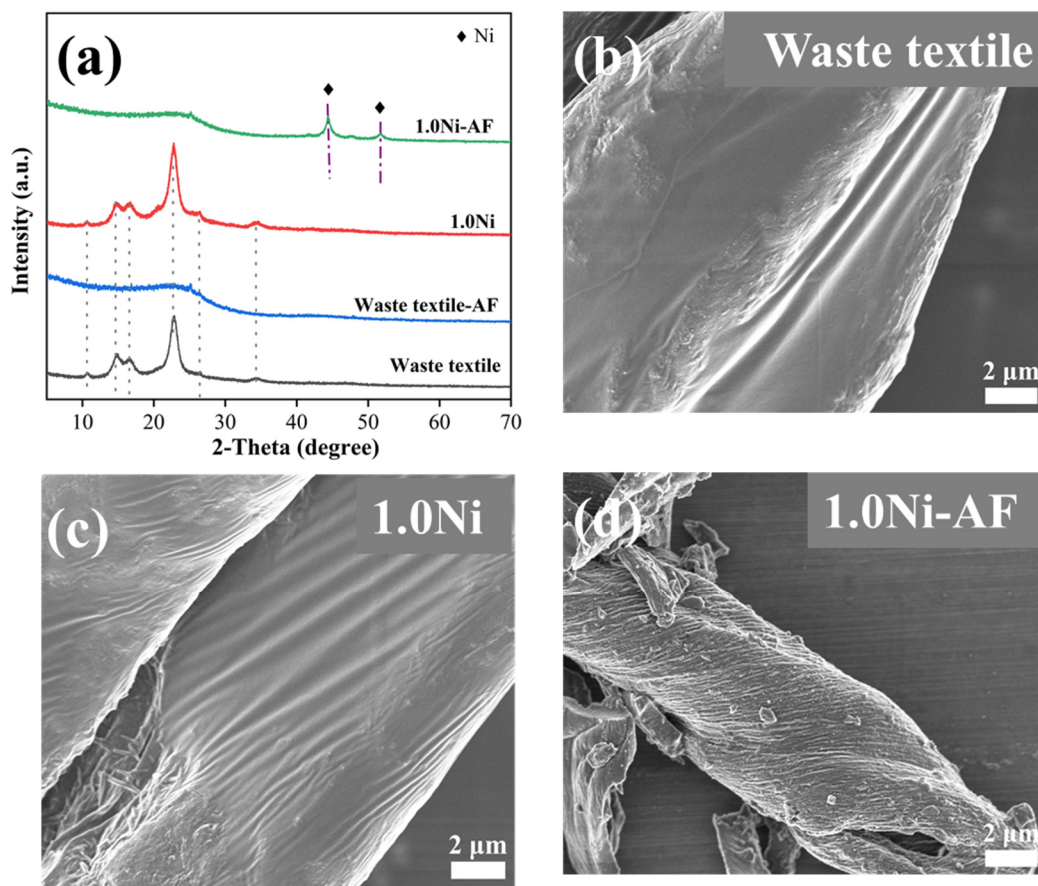


Figure 6. XRD patterns (a) of original waste textile and 1.0Ni waste textile before and after microwave pyrolysis. SEM images of waste textile (b), 1.0Ni (c), and 1.0Ni-AF (d).

A thermogravimetric test (TG) was carried out on the waste textile containing 1.0% Ni to investigate its pyrolysis behavior. As shown in Figure 7a, the waste textile pyrolysis could be divided into three stages. The first stage was the pre-pyrolysis stage, and the temperature range was from 50 to 300 °C. The mass did not change drastically (−4.9%), and the temperature did not reach the decomposition temperature of the material components. There was no drastic pyrolysis, which indicated that the macromolecular organic compounds in the waste textile basically did not start to pyrolyze, but the internal structure of the raw material after warming up changed. At this time, some of the lighter compounds started to react slowly, which made the next step of the raw material mass pyrolysis process easier to carry out. The second stage was the bulk pyrolysis stage in the temperature range from 300 to 526 °C. This process was accompanied by significant weight loss (−76%), which could be found to be the main pyrolysis stage of the waste textile. In this stage, the internal chemical bonds of cellulose and polyester fibers, the main components of the textile, were broken and reconstructed [43]. At the same time, the non-condensable gas-phase products obtained after condensation and separation were also produced in large quantities in this stage. The third stage was the charring stage of the waste textile in the temperature range of 526 °C. The quality was basically unchanged, indicating that the pyrolysis of the internal components of the waste textile was gradually completed [44,45]. According to the TG results, the microwave pyrolysis temperature was controlled at 285 °C, which was lower than the dramatic pyrolysis temperature of the textile (300 °C). It could be seen from the XRD results (Figure 7b) that no decomposition of the textile occurred, but the characteristic peaks attributed to Ni^0 appeared at $2\theta = 45^\circ$, which indicated that Ni^0 was the key to the preparation of hydrogen-containing combustible gas through the microwave pyrolysis of the waste textile.

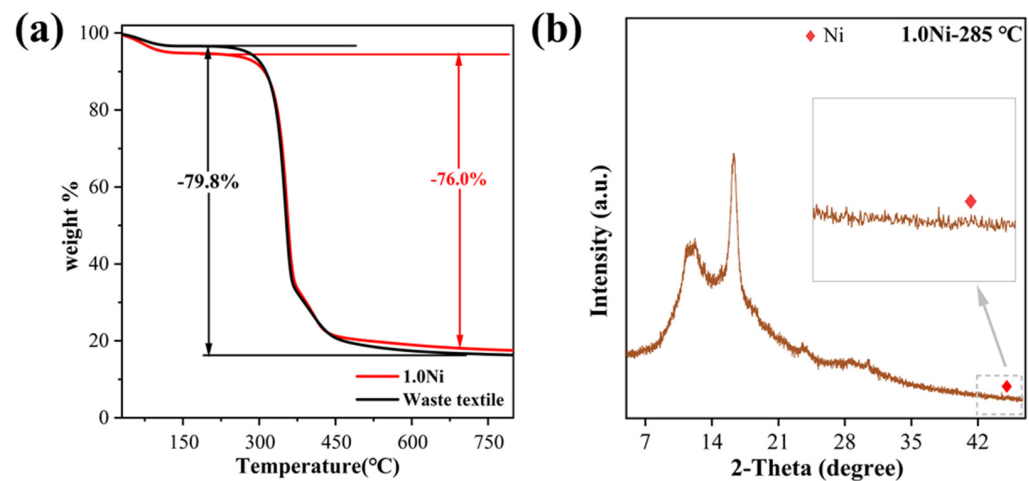


Figure 7. (a) TG curve of waste textile and 1Ni. (b) XRD curves of waste textile containing 1.0Ni after microwave pyrolysis at 285 °C.

3.3. Evaluation of Carbon Emission Reduction and Environmental Implication

In the background of reaching the peak of carbon in 2030 and carbon neutrality in 2060 [46,47], the generation of hydrogen-containing combustible gas from waste textiles impregnated with metal salts using microwave-directed pyrolysis could bring about carbon reduction benefits in the following three areas:

- (1) **Reduce fossil fuel dependence:** The use of waste textiles impregnated with metal salts and microwave-directed pyrolysis to generate hydrogen-containing combustible gas could effectively replace traditional fossil fuels (e.g., coal and oil). Hydrogen, a clean energy source whose combustion product is only water, significantly reduces the emissions of carbon dioxide and other harmful gases. This shift helps slow global warming and reduces air pollution, thereby supporting carbon reduction targets.
- (2) **Improve the resource efficiency of waste textiles:** Treating waste textiles through microwave thermal pyrolysis technology not only effectively disposes of waste textiles and reduces the environmental burden caused by landfills and incineration, but also transforms waste textiles into high-value-added hydrogen and other usable by-products. It improves the resource efficiency of waste materials, reduces carbon emissions during waste treatment, and reduces greenhouse gases at the source.
- (3) **Reduce the carbon footprint of traditional waste treatment:** Traditional waste textile treatment methods such as incineration and landfill usually release a large amount of greenhouse gases such as carbon dioxide. In contrast, the technology of metal salt impregnation and microwave pyrolysis can more efficiently convert waste textiles into hydrogen-containing renewable gases, which reduces the direct greenhouse gas emissions in the waste treatment process. In addition, the higher energy conversion efficiency of this method can further reduce the carbon footprint due to the energy conversion process.

Compared with the traditional conventional pyrolysis method, the microwave method developed in this paper can effectively reduce CO₂ emissions. A set of conventional thermal pyrolysis experiments for the preparation of renewable gas were carried out by placing 1.0Ni-containing materials in a conventional tube furnace for a pyrolysis duration of 5 min and at a pyrolysis temperature of 500 °C. As shown in Figure 8a, the microwave pyrolysis and conventional pyrolysis had about the same solid yield at the same temperature. The liquid yield of microwave pyrolysis decreased significantly, while the pyrolysis capacity was significantly enhanced, and the gas yield increased from 38.0% to 81.1%. This indicated that microwave pyrolysis technology could effectively enhance the pyrolysis efficiency of waste textiles by providing uniform and efficient heating and reduce the energy consumption and greenhouse gas emissions. Further analysis of the gas-phase fractions (Figure 8b) revealed that microwave pyrolysis produced 16.0% more combustible gas than that from

conventional pyrolysis, and its H_2 gas yield was increased by 6.0%, which was 1.6 times higher than that of conventional pyrolysis. It can be seen that microwave pyrolysis can efficiently pyrolyze Ni-containing waste textiles and produce considerable hydrogen-containing combustible gas in a short period of time, which can generate the target products more efficiently and improve the Ni atom utilization rate. Based on the current total generation of low-value solid wastes, if microwave pyrolysis is applied to these wastes, it will be able to significantly reduce the CO_2 emissions during the waste treatment process. Studies have shown that microwave pyrolysis reduces carbon dioxide emissions by about 30~50% compared to those from conventional methods when treating the same amount of waste. This shows that microwave pyrolysis is not only effective in converting waste materials, but also significantly reduces the carbon footprint of the treatment process.

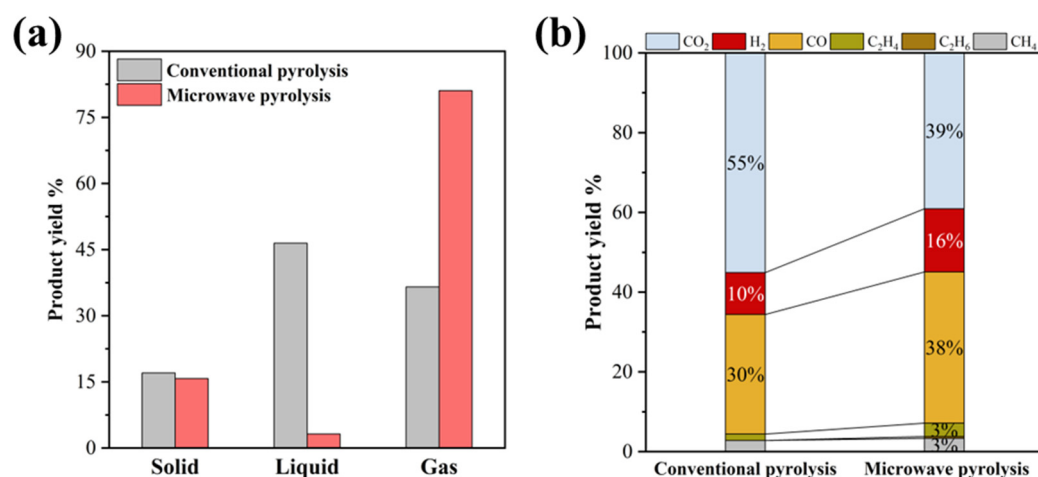


Figure 8. (a) Distribution of products from conventional and microwave pyrolysis of 1.0Ni. (b) Distribution of gas-phase products from conventional and microwave pyrolysis of 1.0Ni.

4. Conclusions

In this study, various types and concentrations of metal salts were loaded onto waste textiles using the impregnation method and subsequently subjected to microwave pyrolysis. The findings reveal that 1.0%Ni achieved the highest hydrogen-rich gas yield of 61.0%, with hydrogen (H_2) production 32.2 times greater than that of the original waste textiles, which was attributed to the uniform dispersion of Ni, promoting the pyrolysis reaction. Compared to conventional pyrolysis, microwave pyrolysis significantly enhanced the gas-phase yield and hydrogen content, while pyrolyzing the waste textiles in a shorter time and effectively reducing CO_2 emissions. The characterization analyses confirmed that Ni^0 acted as an active site during microwave pyrolysis, significantly promoting the pyrolysis of the waste textiles. The results provide practical guidance for solid waste resource management and environmental impact mitigation, thereby advancing the objectives of a green economy and sustainable development.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/pr12112505/s1>, Figure S1: Photo of waste textile; Figure S2: SEM-Mapping of 1.0Ni (a) and 1.0Ni-AF (b); Figure S3: FTIR of dried nickel acetate before and after microwave pyrolysis at 500 °C; Table S1: Element components and element percentage of waste textile.

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References

- Nanda, S.; Berruti, F. A Technical Review of Bioenergy and Resource Recovery from Municipal Solid Waste. *J. Hazard. Mater.* **2021**, *403*, 123970. [\[CrossRef\]](#) [\[PubMed\]](#)
- Subramanian, K.; Chopra, S.S.; Cakin, E.; Li, X.; Lin, C.S.K. Environmental Life Cycle Assessment of Textile Bio-Recycling—Valorizing Cotton-Polyester Textile Waste to Pet Fiber and Glucose Syrup. *Resour. Conserv. Recycl.* **2020**, *161*, 104989. [\[CrossRef\]](#)
- Huang, B.; Zhao, J.; Geng, Y.; Tian, Y.; Jiang, P. Energy-Related GHG Emissions of the Textile Industry in China. *Resour. Conserv. Recycl.* **2017**, *119*, 69–77. [\[CrossRef\]](#)
- Andini, E.; Bhalode, P.; Gantert, E.; Sadula, S.; Vlachos, D.G. Chemical Recycling of Mixed Textile Waste. *Sci. Adv.* **2024**, *10*, eado6827. [\[CrossRef\]](#) [\[PubMed\]](#)
- Fan, W.; Wang, Y.; Liu, R.; Zou, J.; Yu, X.; Liu, Y.; Zhi, C.; Meng, J. Textile Production by Additive Manufacturing and Textile Waste Recycling: A Review. *Environ. Chem. Lett.* **2024**, *22*, 1929–1987. [\[CrossRef\]](#)
- Dong, J.; Tang, Y.; Nzihou, A.; Chi, Y.; Weiss-Hortala, E.; Ni, M.; Zhou, Z. Comparison of Waste-to-Energy Technologies of Gasification and Incineration Using Life Cycle Assessment: Case Studies in Finland, France and China. *J. Clean. Prod.* **2018**, *203*, 287–300. [\[CrossRef\]](#)
- Khan, S.; Anjum, R.; Raza, S.T.; Ahmed Bazai, N.; Ihtisham, M. Technologies for Municipal Solid Waste Management: Current Status, Challenges, and Future Perspectives. *Chemosphere* **2022**, *288*, 132403. [\[CrossRef\]](#)
- Hong, J.; Chen, Y.; Wang, M.; Ye, L.; Qi, C.; Yuan, H.; Zheng, T.; Li, X. Intensification of Municipal Solid Waste Disposal in China. *Renew. Sustain. Energy Rev.* **2017**, *69*, 168–176. [\[CrossRef\]](#)
- Cao, C.; Cheng, Y.; Hu, H.; Wang, H.; Liu, S.; Hu, M.; Li, X.; Yao, H. Products Distribution and Sulfur Fixation during the Pyrolysis of CaO Conditioned Textile Dyeing Sludge: Effects of Pyrolysis Temperature and Heating Rate. *Waste Manag.* **2022**, *153*, 367–375. [\[CrossRef\]](#)
- Yin, M.; Bi, D.; Zhao, W.; Liu, J.; Zhao, A.; Jiang, M. Production of Bio-Oil and Biochar by the Nitrogen-Rich Pyrolysis of Cellulose with Urea: Pathway of Nitrile in Bio-Oil and Evolution of Nitrogen in Biochar. *J. Anal. Appl. Pyrolysis* **2023**, *174*, 106137. [\[CrossRef\]](#)
- Zhu, Y.; Huo, G.; Yang, W.; Liu, H.; Zhang, W.; Cheng, W.; Yang, H.; Wang, Z.; Jin, Y.; Zhao, H. Catalytic Pyrolysis of Duckweed with Phosphoric Acid: Pyrolysis Behavior and Kinetics Analysis. *J. Anal. Appl. Pyrolysis* **2024**, *177*, 106384. [\[CrossRef\]](#)
- Ren, X.; Shanb Ghazani, M.; Zhu, H.; Ao, W.; Zhang, H.; Moreside, E.; Zhu, J.; Yang, P.; Zhong, N.; Bi, X. Challenges and Opportunities in Microwave-Assisted Catalytic Pyrolysis of Biomass: A Review. *Appl. Energy* **2022**, *315*, 118970. [\[CrossRef\]](#)
- Ge, S.; Yek, P.N.Y.; Cheng, Y.W.; Xia, C.; Wan Mahari, W.A.; Liew, R.K.; Peng, W.; Yuan, T.-Q.; Tabatabaei, M.; Aghbashlo, M.; et al. Progress in Microwave Pyrolysis Conversion of Agricultural Waste to Value-Added Biofuels: A Batch to Continuous Approach. *Renew. Sustain. Energy Rev.* **2021**, *135*, 110148. [\[CrossRef\]](#)
- Foong, S.Y.; Liew, R.K.; Yang, Y.; Cheng, Y.W.; Yek, P.N.Y.; Wan Mahari, W.A.; Lee, X.Y.; Han, C.S.; Vo, D.-V.N.; Van Le, Q.; et al. Valorization of Biomass Waste to Engineered Activated Biochar by Microwave Pyrolysis: Progress, Challenges, and Future Directions. *Chem. Eng. J.* **2020**, *389*, 124401. [\[CrossRef\]](#)
- Huang, Y.F.; Kuan, W.H.; Lo, S.L.; Lin, C.F. Total Recovery of Resources and Energy from Rice Straw Using Microwave-Induced Pyrolysis. *Bioresour. Technol.* **2008**, *99*, 8252–8258. [\[CrossRef\]](#)
- Parvez, A.M.; Wu, T.; Afzal, M.T.; Mareta, S.; He, T.; Zhai, M. Conventional and Microwave-Assisted Pyrolysis of Gumwood: A Comparison Study Using Thermodynamic Evaluation and Hydrogen Production. *Fuel Process. Technol.* **2019**, *184*, 1–11. [\[CrossRef\]](#)
- Kim, D.; Kim, G.; Oh, D.Y.; Seong, K.-W.; Park, K.Y. Enhanced Hydrogen Production from Anaerobically Digested Sludge Using Microwave Assisted Pyrolysis. *Fuel* **2022**, *314*, 123091. [\[CrossRef\]](#)
- Freitas, L.C.; Barbosa, J.R.; da Costa, A.L.C.; Bezerra, F.W.F.; Pinto, R.H.H.; Carvalho Junior, R.N. de From Waste to Sustainable Industry: How Can Agro-Industrial Wastes Help in the Development of New Products? *Resour. Conserv. Recycl.* **2021**, *169*, 105466. [\[CrossRef\]](#)
- Dai, L.; Wang, Y.; Liu, Y.; He, C.; Ruan, R.; Yu, Z.; Jiang, L.; Zeng, Z.; Wu, Q. A Review on Selective Production of Value-Added Chemicals via Catalytic Pyrolysis of Lignocellulosic Biomass. *Sci. Total Environ.* **2020**, *749*, 142386. [\[CrossRef\]](#)
- Zhuang, X.; Zhu, N.; Li, F.; Lin, H.; Liang, C.; Dang, Z.; Zou, Y. Hydrogen-Rich Syngas Production from Waste Textile Gasification Coupling with Catalytic Reforming under Steam Atmosphere. *Processes* **2024**, *12*, 1790. [\[CrossRef\]](#)
- Kwon, D.; Yi, S.; Jung, S.; Kwon, E.E. Valorization of Synthetic Textile Waste Using CO₂ as a Raw Material in the Catalytic Pyrolysis Process. *Environ. Pollut.* **2021**, *268*, 115916. [\[CrossRef\]](#) [\[PubMed\]](#)
- Yao, D.; Li, H.; Dai, Y.; Wang, C.-H. Impact of Temperature on the Activity of Fe-Ni Catalysts for Pyrolysis and Decomposition Processing of Plastic Waste. *Chem. Eng. J.* **2021**, *408*, 127268. [\[CrossRef\]](#)

23. Ren, L.; Zhang, K.; Zhang, Y.; Wang, F.; Yang, F.; Cheng, F. Mechanism of Gas Production under Microwave/Conventional Pyrolysis of Sewage Sludge: Mechanism of Microwave Energy Action on Oxygen-Containing Functional Groups. *Chem. Eng. J.* **2023**, *464*, 142511. [\[CrossRef\]](#)
24. Wang, X.; Wang, C.; Chen, Z.; Liu, B.; Ma, W.; Tang, Z.; Chen, Y.; Yang, H.; Chen, H. Study on the Effect Mechanism of Ca and Mg with Different Anion on Lignocellulosic Biomass Pyrolysis. *J. Anal. Appl. Pyrolysis* **2024**, *177*, 106335. [\[CrossRef\]](#)
25. Baidya, T.; Cattolica, R.J.; Seiser, R. High Performance Ni-Fe-Mg Catalyst for Tar Removal in Producer Gas. *Appl. Catal. A Gen.* **2018**, *558*, 131–139. [\[CrossRef\]](#)
26. Lu, Q.; Yuan, S.; Li, J.; Chen, X.; Li, K.; Xie, X.; Fu, X.; He, Z. Influence of Mg and Co Addition on Fe Based Catalyst for In-Situ Biomass Pyrolysis. *J. Anal. Appl. Pyrolysis* **2023**, *169*, 105832. [\[CrossRef\]](#)
27. Patwardhan, P.R.; Satrio, J.A.; Brown, R.C.; Shanks, B.H. Influence of Inorganic Salts on the Primary Pyrolysis Products of Cellulose. *Bioresour. Technol.* **2010**, *101*, 4646–4655. [\[CrossRef\]](#)
28. Shang, S.; Guo, C.; Lan, K.; Li, Z.; He, W.; Qin, Z.; Li, J. Hydrogen-Rich Syngas Production via Catalytic Gasification of Sewage Sludge and Wheat Straw Using Corn Stalk Char-Supported Catalysts. *BioResources* **2020**, *15*, 4294–4313. [\[CrossRef\]](#)
29. Yousef, S.; Eimontas, J.; Striugas, N.; Tatarants, M.; Abdelnaby, M.A.; Tuckute, S.; Kliucininkas, L. A Sustainable Bioenergy Conversion Strategy for Textile Waste with Self-Catalysts Using Mini-Pyrolysis Plant. *Energy Convers. Manag.* **2019**, *196*, 688–704. [\[CrossRef\]](#)
30. Eskandari, S.; Tate, G.; Leaphart, N.R.; Regalbuto, J.R. Nanoparticle Synthesis via Electrostatic Adsorption Using Incipient Wetness Impregnation. *ACS Catal.* **2018**, *8*, 10383–10391. [\[CrossRef\]](#)
31. Stoyanova, M.; Schneider, M.; Pohl, M.M.; Rodemerck, U. Direct Conversion of Ethylene to Propene over Ni Impregnated by Incipient Wetness on Silica-Alumina. *Catal. Commun.* **2017**, *92*, 65–69. [\[CrossRef\]](#)
32. Chen, S.; Wang, T.; Song, W.; Tang, Z.; Cao, Z.; Chen, H.; Lian, Y.; Hu, X.; Zheng, W.; Lian, H. A Novel Particulate Matter Sampling and Cell Exposure Strategy Based on Agar Membrane for Cytotoxicity Study. *Chemosphere* **2022**, *300*, 134473. [\[CrossRef\]](#) [\[PubMed\]](#)
33. Liu, H.; Li, N.; Feng, M.; Li, G.; Zhang, W.; An, T. Near-Infrared Light Induced Adsorption–Desorption Cycle for VOC Recovery by Integration of Metal–Organic Frameworks with Graphene Oxide Nanosheets. *Environ. Sci. Nano* **2022**, *9*, 1858–1868. [\[CrossRef\]](#)
34. Meshksar, M.; Farsi, M.; Rahimpour, M.R. Effect of Ni Active Site Position and Synthesis Route on Activity, Stability, and Morphology of Ce Promoted Ni/Al₂O₃ Catalyst for Clean H₂ Production. *J. Environ. Chem. Eng.* **2022**, *10*, 108471. [\[CrossRef\]](#)
35. Yao, D.; Zhang, Y.; Williams, P.T.; Yang, H.; Chen, H. Co-Production of Hydrogen and Carbon Nanotubes from Real-World Waste Plastics: Influence of Catalyst Composition and Operational Parameters. *Appl. Catal. B Environ.* **2018**, *221*, 584–597. [\[CrossRef\]](#)
36. Ochoa, A.; Barbarias, I.; Artetxe, M.; Gayubo, A.G.; Olazar, M.; Bilbao, J.; Castaño, P. Deactivation Dynamics of a Ni Supported Catalyst during the Steam Reforming of Volatiles from Waste Polyethylene Pyrolysis. *Appl. Catal. B Environ.* **2017**, *209*, 554–565. [\[CrossRef\]](#)
37. Hwang, H.; Oh, S.; Cho, T.S.; Choi, I.G.; Choi, J.W. Fast Pyrolysis of Potassium Impregnated Poplar Wood and Characterization of Its Influence on the Formation as Well as Properties of Pyrolytic Products. *Bioresour. Technol.* **2013**, *150*, 359–366. [\[CrossRef\]](#)
38. Wang, W.; Lemaire, R.; Bensakhria, A.; Luat, D. Journal of Analytical and Applied Pyrolysis Review on the Catalytic Effects of Alkali and Alkaline Earth Metals (AAEMs) Including Sodium, Potassium, Calcium and Magnesium on the Pyrolysis of Lignocellulosic Biomass and on the Co-Pyrolysis of Coal With. *J. Anal. Appl. Pyrolysis* **2022**, *163*, 105479. [\[CrossRef\]](#)
39. Jia, X.; Wei, Y.; Si, R.; Wang, C.; Guo, P.; Ji, J. Catalytic Production of H₂-Rich Syngas from Cellulose Pyrolysis under Nickel Metal Fog with Molten Carbonates. *Int. J. Hydrogen Energy* **2024**, *72*, 991–1000. [\[CrossRef\]](#)
40. Buffoni, I.N.; Pompeo, F.; Santori, G.F.; Nichio, N.N. Nickel Catalysts Applied in Steam Reforming of Glycerol for Hydrogen Production. *Catal. Commun.* **2009**, *10*, 1656–1660. [\[CrossRef\]](#)
41. Choya, A.; de Rivas, B.; No, M.L.; Gutiérrez-Ortiz, J.I.; López-Fonseca, R. Dry Reforming of Methane over Sub-Stoichiometric NiAl₂O₄-Mediated Ni/Al₂O₃ Catalysts. *Fuel* **2024**, *358*, 130166. [\[CrossRef\]](#)
42. Du, J.; Gao, J.; Gu, F.; Zhuang, J.; Lu, B.; Jia, L.; Xu, G.; Liu, Q.; Su, F. A Strategy to Regenerate Coked and Sintered Ni/Al₂O₃ Catalyst for Methanation Reaction. *Int. J. Hydrogen Energy* **2018**, *43*, 20661–20670. [\[CrossRef\]](#)
43. Rubi, R.V.C.; Allayban, J.P.O.; Deduque, D.A.B.; Mena, J.A.B.; Robles, N.M.D.; Rubio, B.D.; Roque, E.C.; Joy Janaban, P.; Olay, J.G. Production and Characterization of Activated Carbon from Pyrolysis Biochar of Cellulosic Cotton-Based Textile Wastes. *Mater. Today Proc.* **2023**. [\[CrossRef\]](#)
44. Yousef, S.; Eimontas, J.; Zakarauskas, K.; Striugas, N.; Mohamed, A. A New Strategy for Using Lint-Microfibers Generated from Clothes Dryer as a Sustainable Source of Renewable Energy. *Sci. Total Environ.* **2021**, *762*, 143107. [\[CrossRef\]](#)
45. Ling, C.; Shi, S.; Hou, W.; Yan, Z. Separation of Waste Polyester/Cotton Blended Fabrics by Phosphotungstic Acid and Preparation of Terephthalic Acid. *Polym. Degrad. Stab.* **2019**, *161*, 157–165. [\[CrossRef\]](#)
46. Dinga, C.D.; Wen, Z. China's Green Deal: Can China's Cement Industry Achieve Carbon Neutral Emissions by 2060? *Renew. Sustain. Energy Rev.* **2022**, *155*, 111931. [\[CrossRef\]](#)
47. Zhang, S.-C.; Yang, X.-Y.; Xu, W.; Fu, Y.-J. Contribution of Nearly-Zero Energy Buildings Standards Enforcement to Achieve Carbon Neutral in Urban Area by 2060. *Adv. Clim. Chang. Res.* **2021**, *12*, 734–743. [\[CrossRef\]](#)

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